

# Minseo Kim (김민서)

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## Education

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### POSTECH (Pohang University of Science and Technology)

*Feb 2022 – Present*

B.S. in Computer Science and Engineering

## Work Experience

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### ENERZAI Inc.

*Jul 2026 – Present*

#### AI Compiler Engineer

Developing a compiler for an in-house programming language targeting high-performance AI workloads on edge devices.

- Alternative Military Service (산업기능요원)
- Skills: MLIR, C++

### Wordbricks LLC

*Jun 2025 – Aug 2025*

#### Full-stack Web Developer Intern

Led and developed an LLM-based web crawling API and dashboard as a Product Owner.

- Key Contributions:
  - Planned and launched the service.
  - Designed a web crawling API.
  - Developed the API dashboard and documentation website.
- Skills: Next.js, Hono, Cloudflare Workers, Vercel AI SDK, Prisma, Supabase

### POSTECH Data Systems Lab

*Jan 2023 – Jan 2024*

#### Undergraduate Researcher

Led the end-to-end development of an integrated database website for an academic conference.

- Key Contributions:
  - Developed backend and frontend.
  - Designed a data integration pipeline.
  - Built real-time data visualization and optimized database queries.
- Skills: React.js, Express.js, PostgreSQL, Airflow, Docker, Nginx

## Personal Projects

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### Tiny ML Compiler

*May 2025*

Compiler for Tiny ML, a functional programming language based on Standard ML.

- Key Implementations:
  - Hindley-Milner type system based on Algorithm W, featuring let-polymorphism.
  - Code generation to the assembly language of a virtual machine.
  - Support for algebraic data types (ADTs), including recursive types and pattern matching.
  - Support for closures, higher-order functions, and recursive functions.
- Skills: OCaml

## Pintos

Sep 2025 – Dec 2025

Educational operating system for 80x86 architecture (based on Stanford CS140)

- Key Implementations:
  - Kernel-level threading with priority scheduling and synchronization primitives (locks, semaphores).
  - System call implementation for user program execution, process management, and file operations.
  - Virtual memory system featuring demand paging, stack growth, and memory-mapped files.
- Skills: C, x86 Assembly, GDB

## RISC-V CPU

Mar 2025 – May 2025

5-stage pipelined RISC-V CPU (2-person project)

- Implemented a subset of the RV32I ISA.
- Key Implementations:
  - Hazard detection unit with data forwarding.
  - Gshare branch predictor with a 2-bit saturating counter and pattern history table (PHT).
  - Direct-mapped cache.
- Skills: Verilog HDL

## Sponge

Mar 2025 – May 2025

Educational network stack development project (based on Stanford CS144)

- Key Implementations:
  - TCP flow control.
  - Routing table lookup and packet forwarding.
  - ARP and IP packet processing.
- Skills: C++

## BaroKey

Nov 2023

Submitted to the 3rd UniDThon Hackathon (team project).

- Developed a web service that provides users with real-time emergency keywords, such as safety threats or natural disasters, based on their location.
- Role: Team Leader, Full-stack Developer (Led Backend & Infrastructure)

## Skills

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|-----------------------|---------------------------------------------|
| Programming Languages | C, C++, Python, TypeScript, Java, Go, OCaml |
| System                | Linux, FreeBSD, GDB                         |
| Web Development       | React.js, Next.js, Express.js, FastAPI      |
| Database              | PostgreSQL                                  |
| Infrastructure        | Docker, Nginx                               |